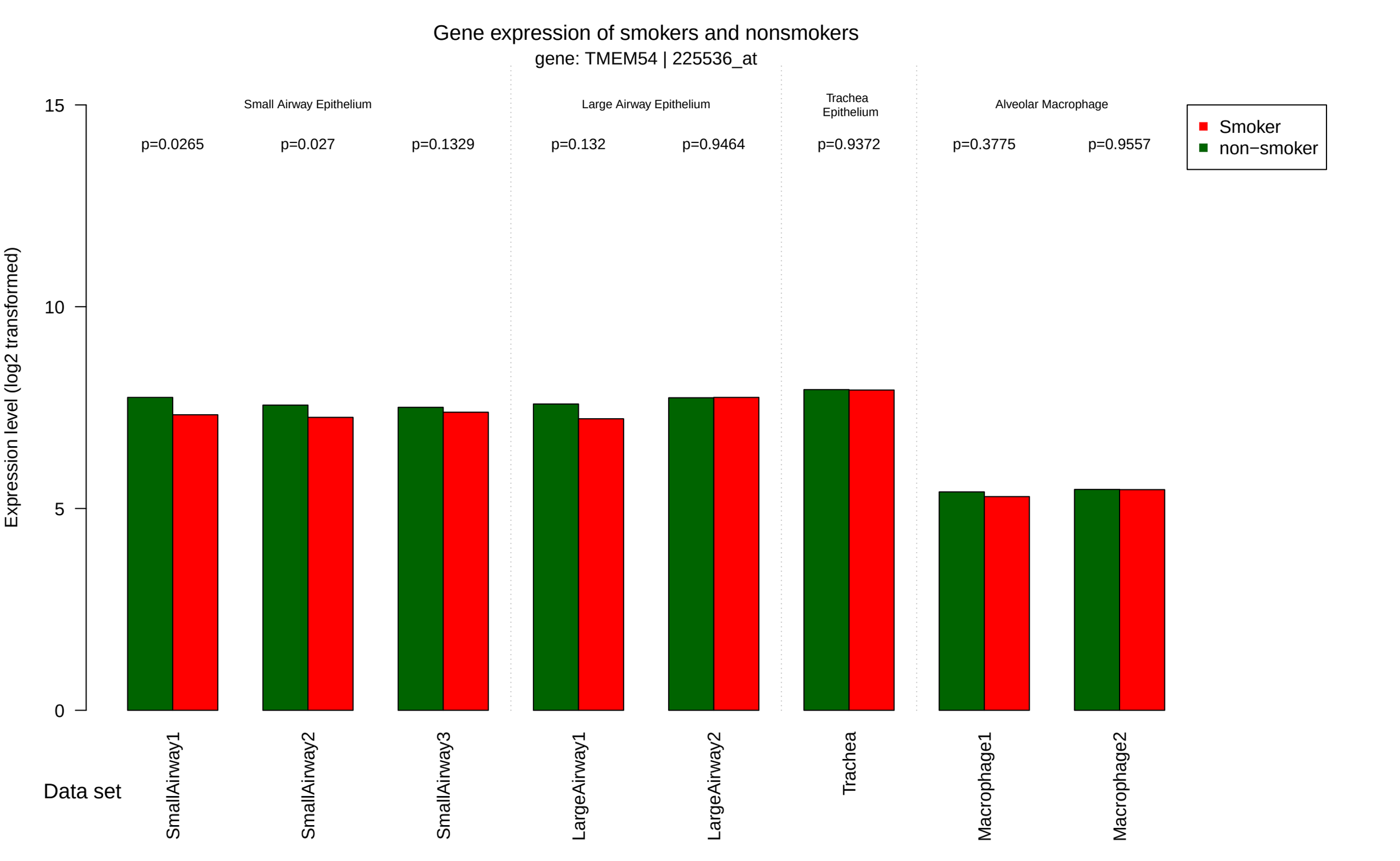
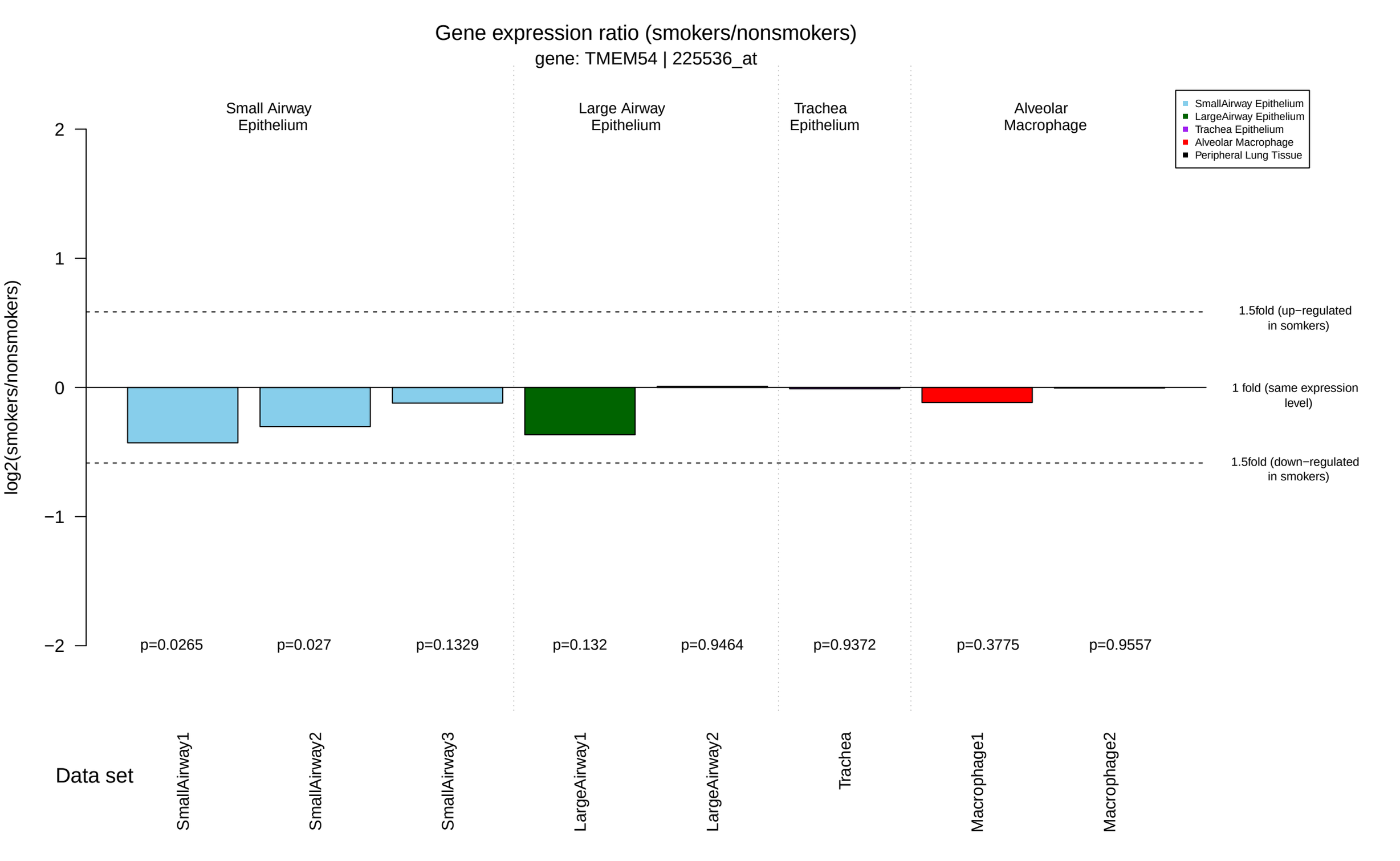
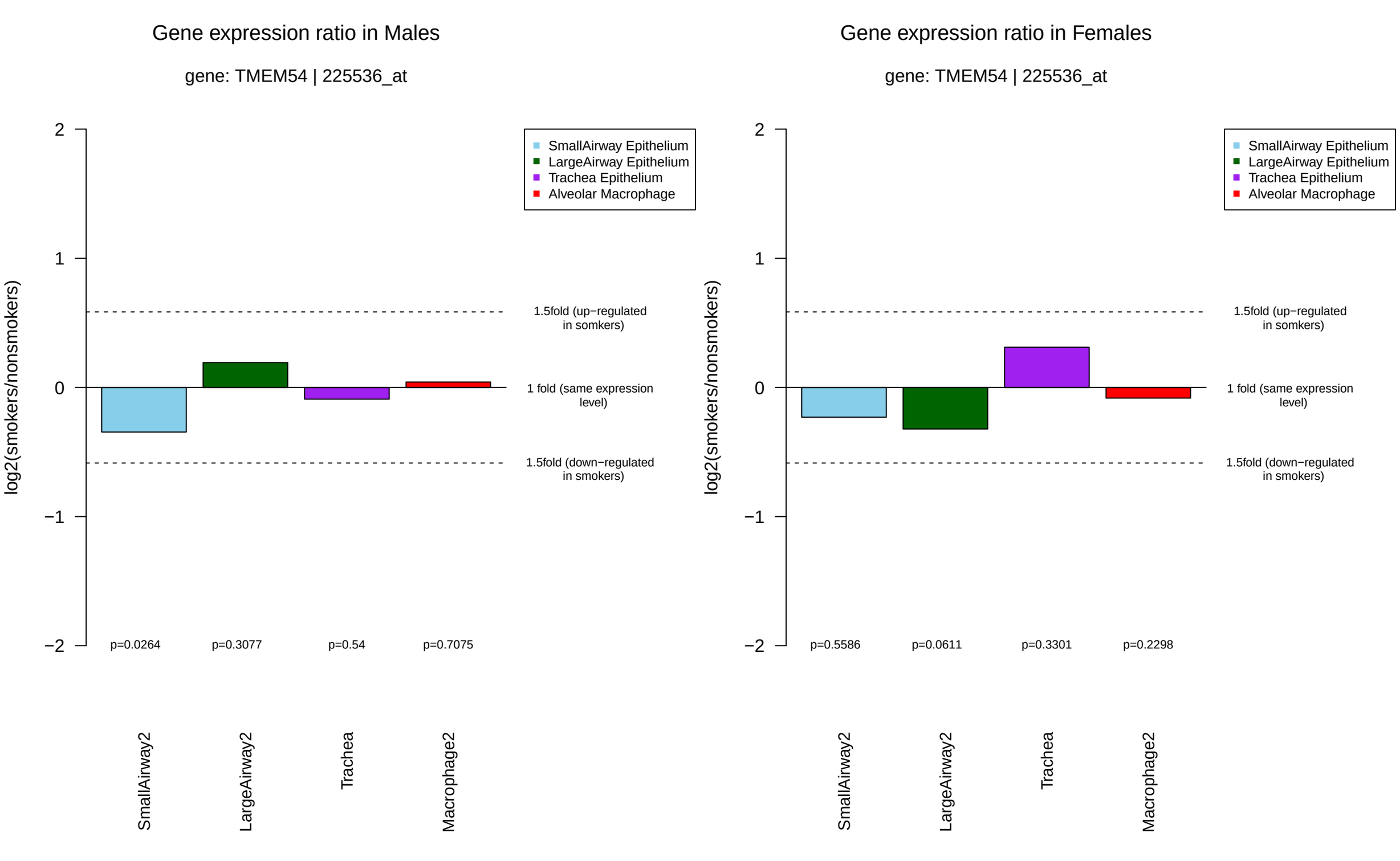
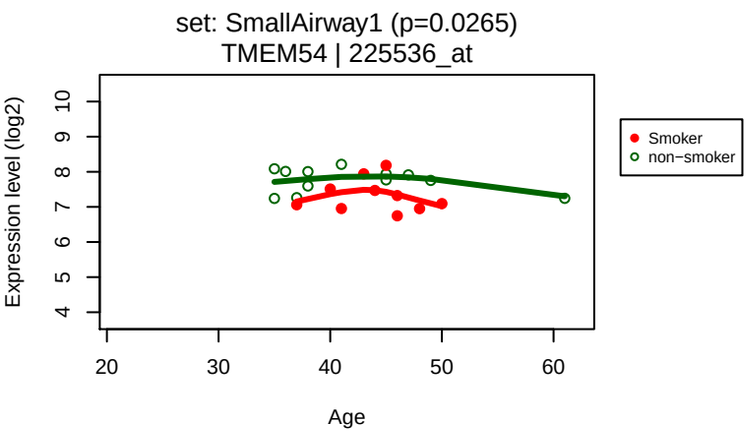
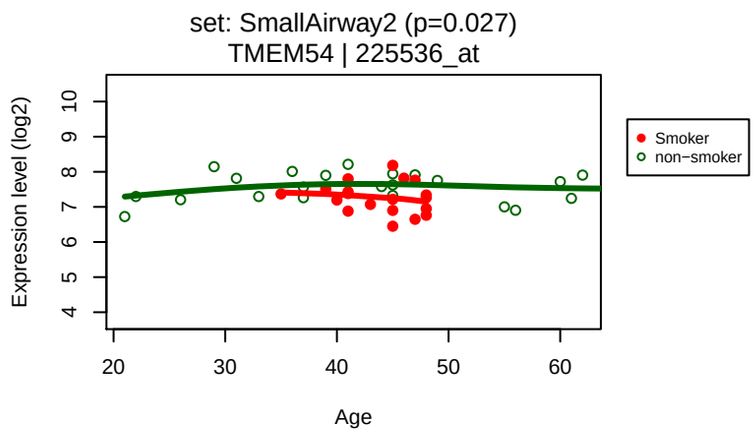
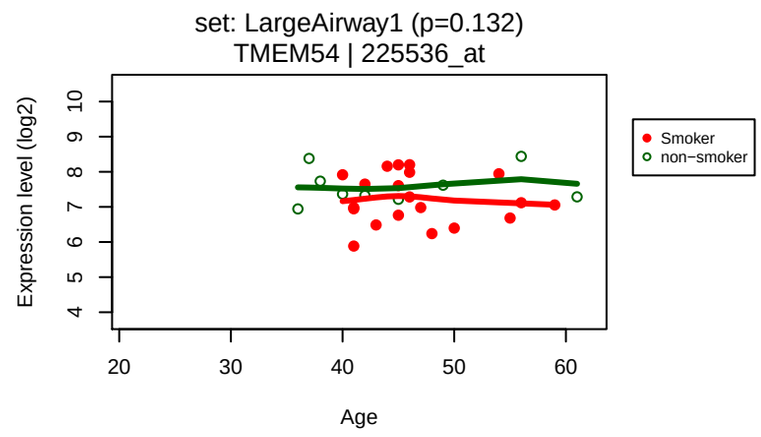

Gene expression of smokers and nonsmokers
gene: TMEM54 | 225536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0265
p=0.027
p=0.1329
p=0.132
p=0.9464
p=0.9372
p=0.3775
p=0.9557
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

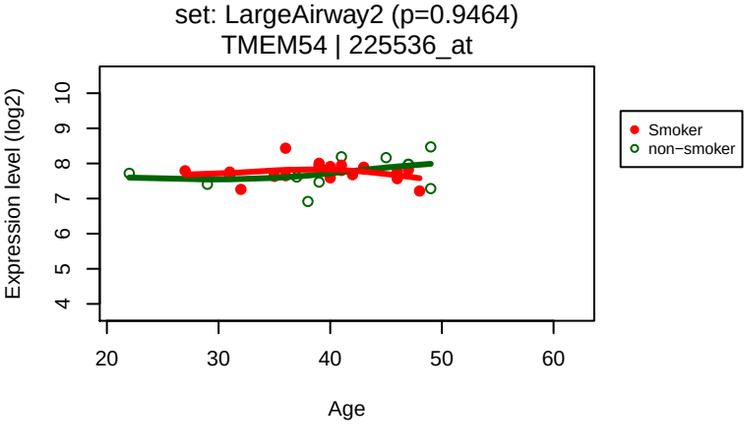

Gene expression ratio (smokers/nonsmokers)
gene: TMEM54 | 225536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0265
p=0.027
p=0.1329
p=0.132
p=0.9464
p=0.9372
p=0.3775
p=0.9557
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

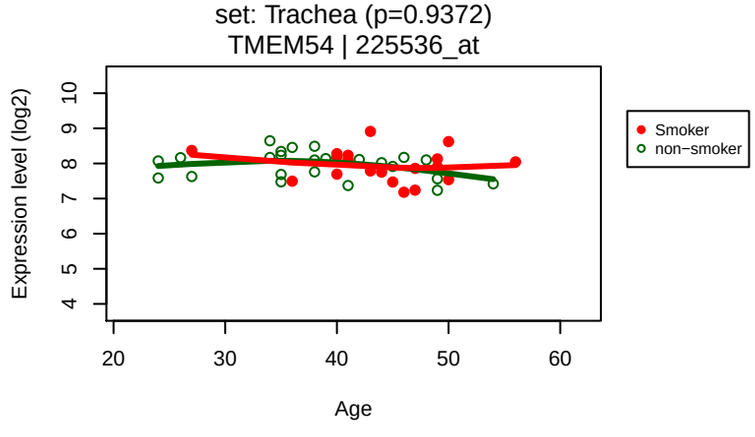

Gene expression ratio in Males
gene: TMEM54 | 225536_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0264
p=0.3077
p=0.54
p=0.7075
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM54 | 225536_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5586
p=0.0611
p=0.3301
p=0.2298
SmallAirway2
LargeAirway2
Trachea
Macrophage2

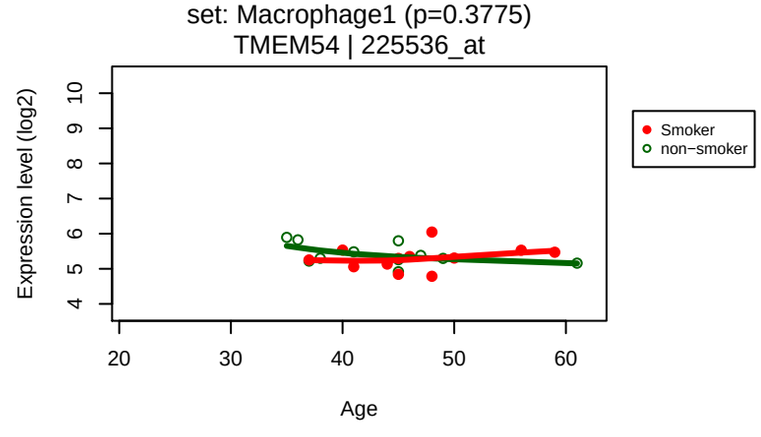

set: SmallAirway1 (p=0.0265)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker

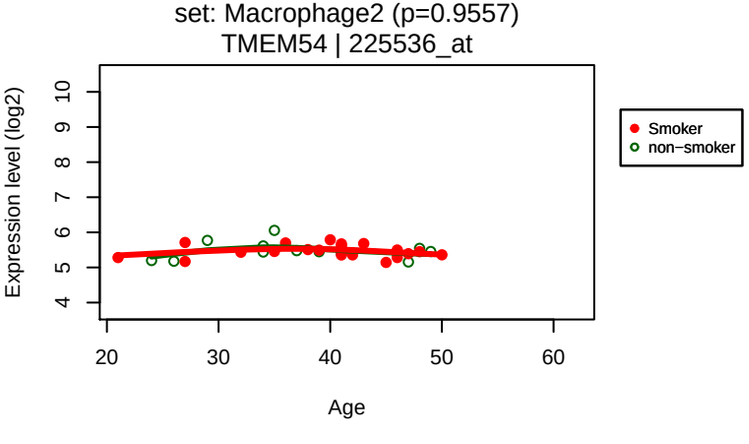

set: SmallAirway2 (p=0.027)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker

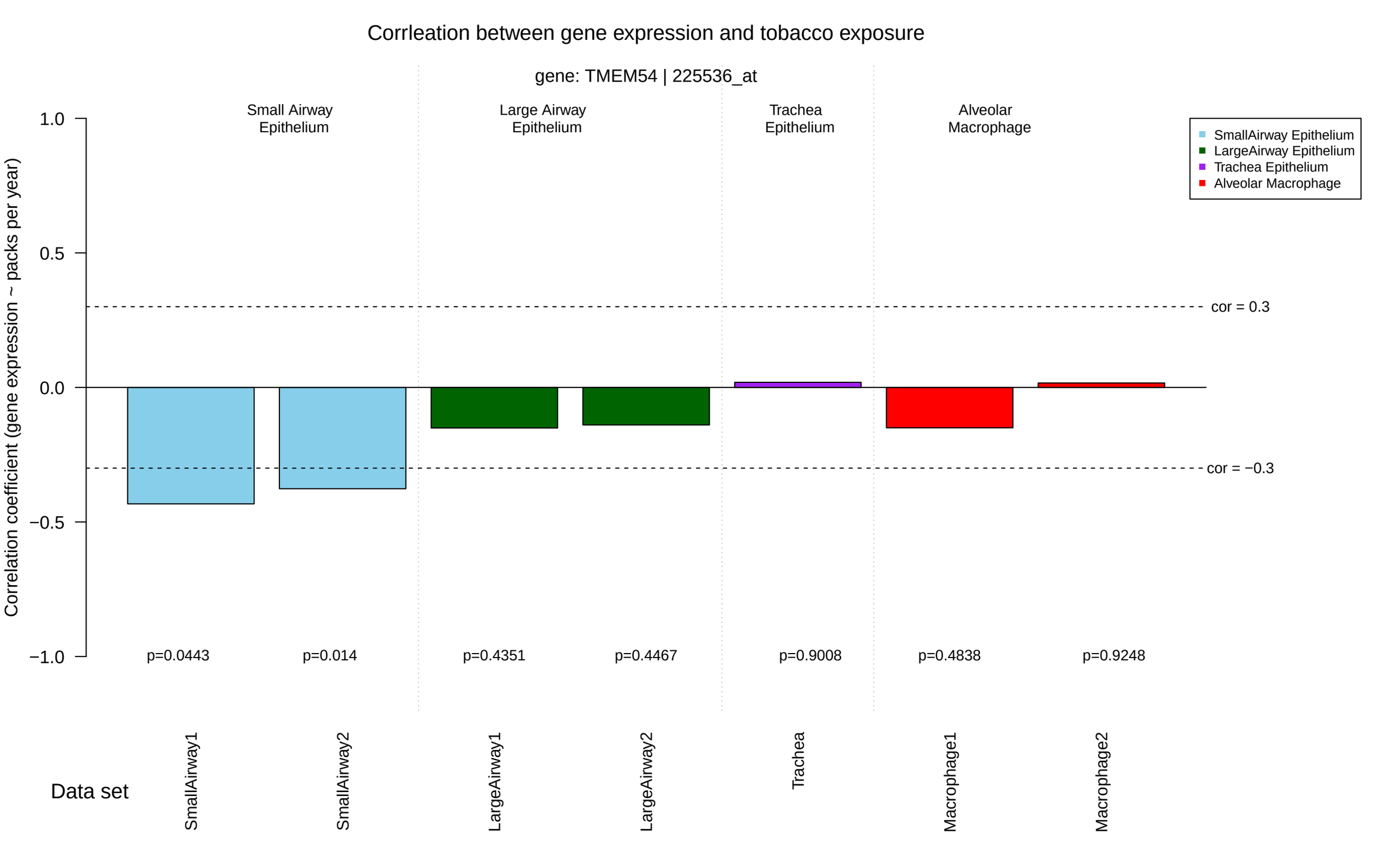

set: LargeAirway1 (p=0.132)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9464)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9372)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3775)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9557)
TMEM54 | 225536_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM54 | 225536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0443
p=0.014
p=0.4351
p=0.4467
p=0.9008
p=0.4838
p=0.9248
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2